

The Twin Paradox

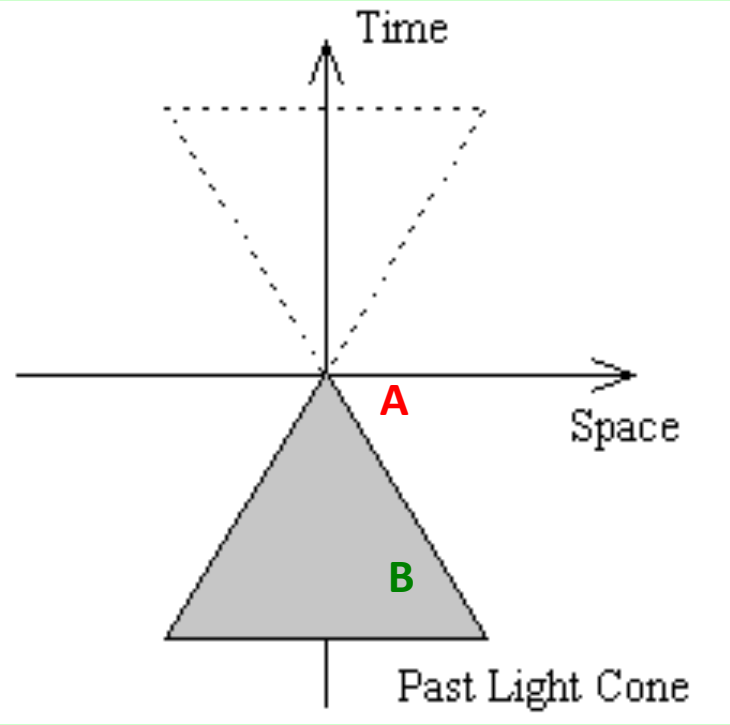
Group Discussion

The “Butterfly Effect”: things far away can effect you in large and unpredictable ways.

The figure to the right has 2 events. A is your roommate shooting a NERF missile at you just moments ago, while **B** is an alien in Andromeda brushing his fur millions of years ago.

Which could affect on your life?

What do you think ultimately influences us more: the weak/distant or the near/strong?



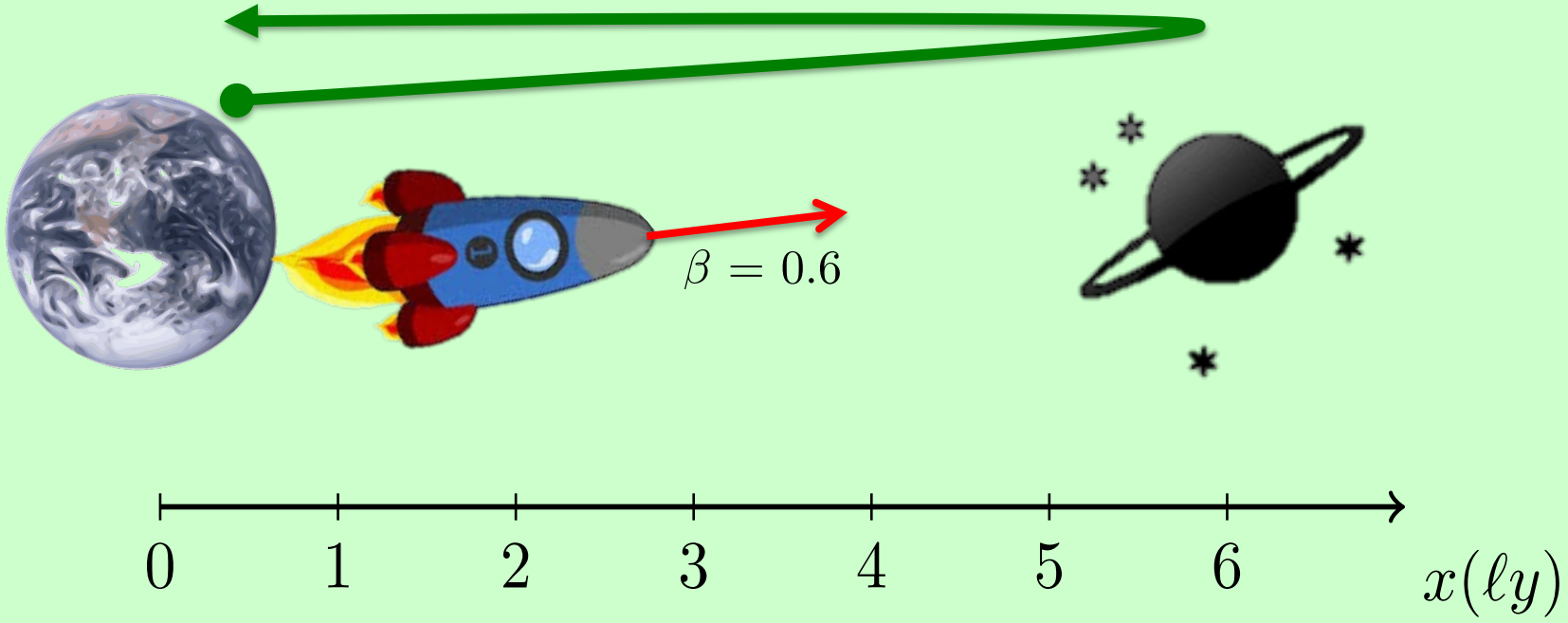


Which Path is Shorter?

Golden rule – geometry of spacetime is not the same as that of a sheet of paper --- the shortest distance between two events is *not* a straight line.

Twin Paradox

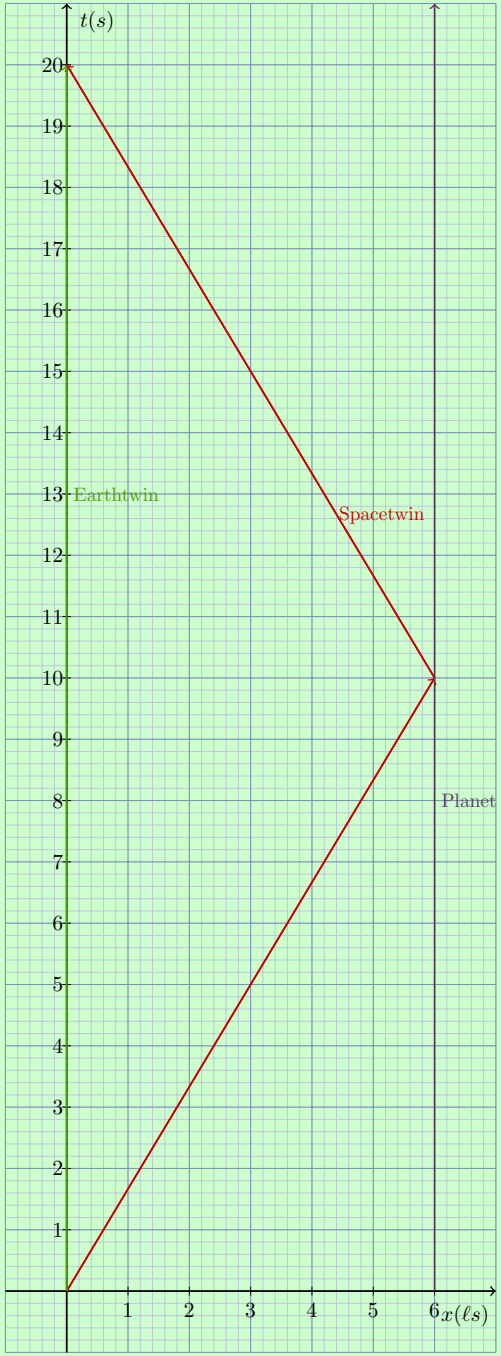
Earthtwin stays home
Spacetwin travels 6 ly at $\beta = 0.6$ to a planet then returns home
Who ages more?



Spacetime diagram:

Think in-terms of worldline

(The proper time is the length
Of the world line.)



Viewpoint #1: Propertime is the length of the worldline

$$\Delta S^2 = -\Delta t^2 + \Delta x^2$$

$$\Delta S = \sqrt{-\Delta S^2}$$

$$\tau = \Delta S = \sqrt{t^2 - x^2}$$

For complex reasons

For any piecewise straight worldline just break up

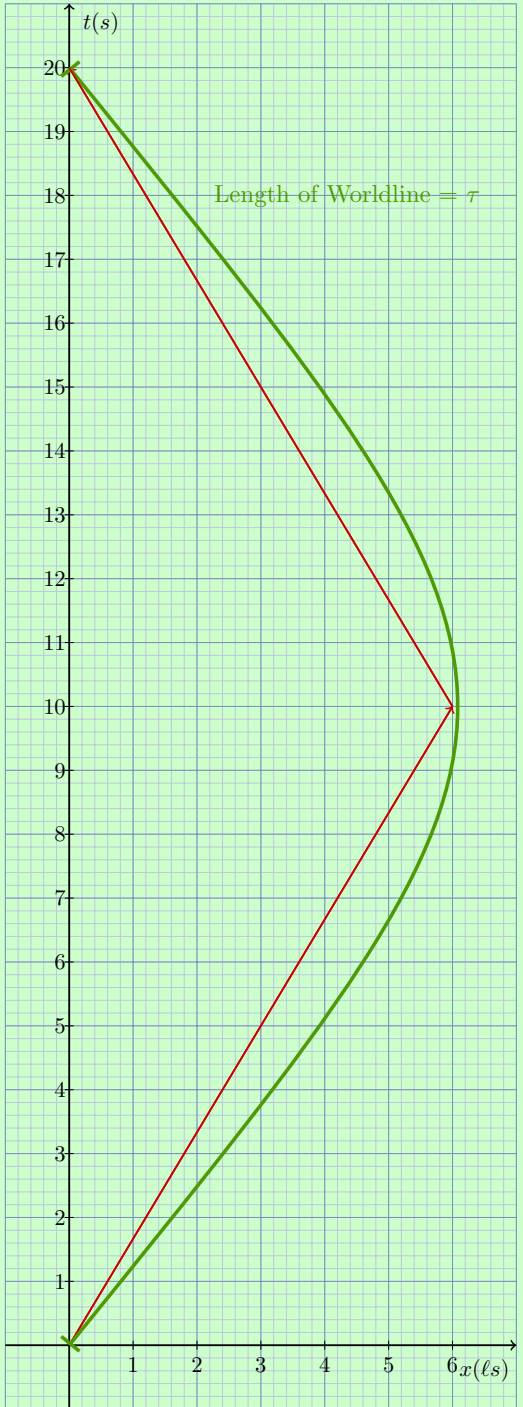
Earthtwin:

$$\tau = \Delta S = \sqrt{t^2 - x^2} = 2 \times \sqrt{20^2 - 0^2} = \boxed{20 \text{ yrs}}$$

Spacetwin:

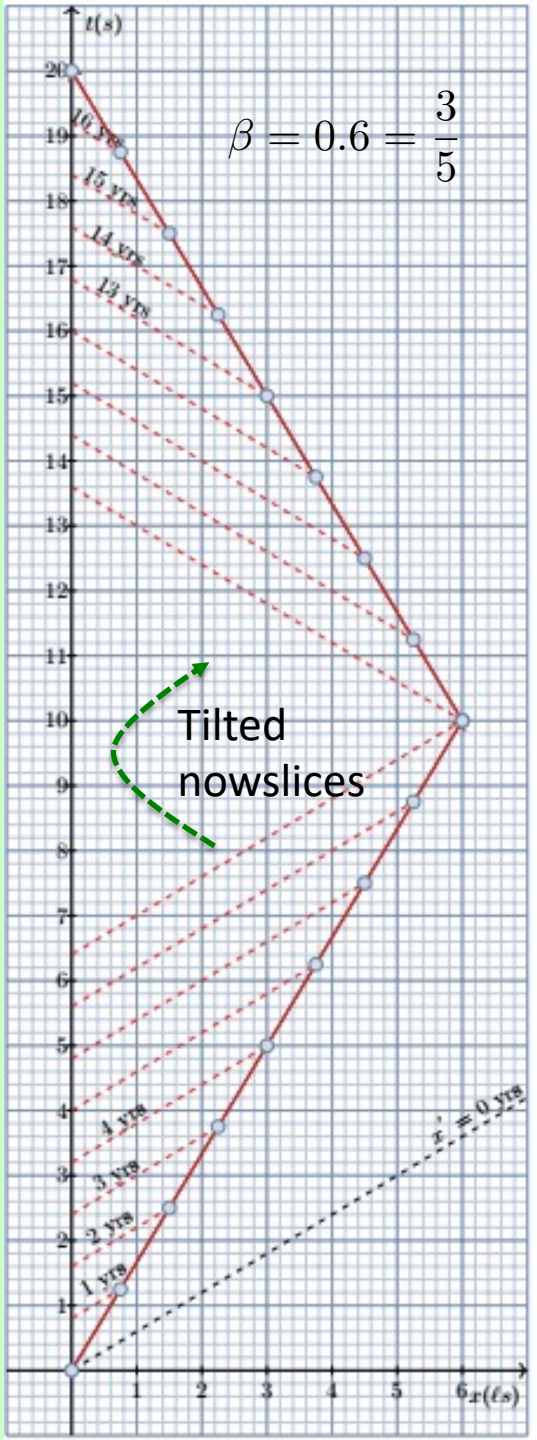
$$\begin{aligned} \tau &= \Delta S_{\text{outbound}} + \Delta S_{\text{inbound}} = 2 \times \Delta S_{\text{out}} \\ &= 2\sqrt{t_{\text{out}}^2 - x_{out}^2} = 2 \times \sqrt{10^2 - 6^2} = \boxed{16 \text{ yrs}} \end{aligned}$$

Spacetwin should be 4 years younger when they are reunited.



Viewpoint #2: Tilted Now-Slices

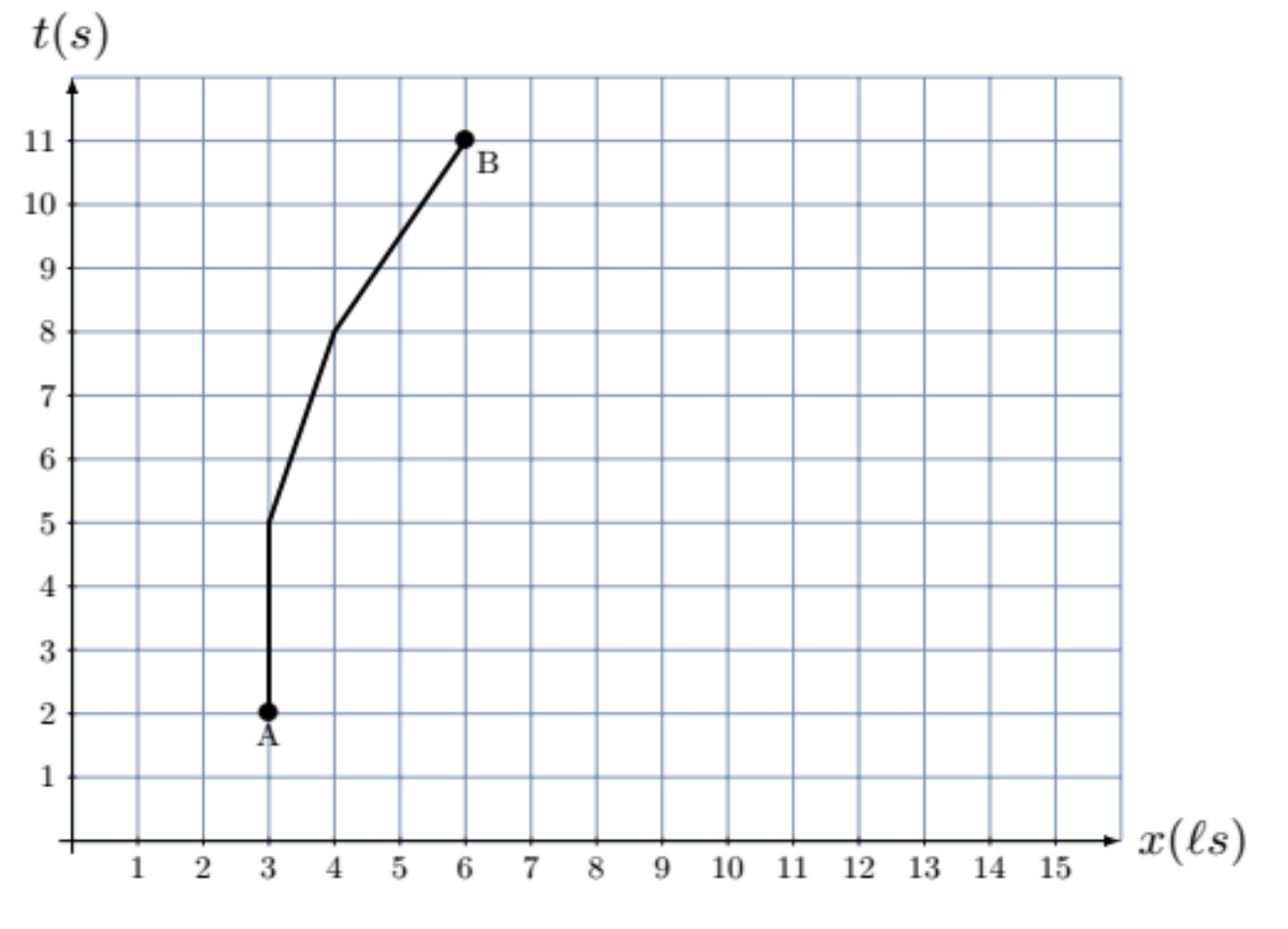
1. Spacetwin fires once per year.
2. Spacetwin now-slices have slope: $m = \beta = 0.6 = \frac{3}{5}$, tilted away from Earthtwin.
3. When Spacetwin arrives at distant planet. Spaceship **accelerates** to stop and then start again.
4. Spaceship now-slices now have slope: $m = \beta = 0.6 = \frac{3}{5}$, tilted toward Earthtwin.
5. The acceleration changes which Earthtwin and Spacetwin are paired together.
6. **Acceleration Tilts Now – Slices.**



Ex. Kwame takes the following trip

A. Find Δs between event A and B. Is the separation time, space, or lightlike?

B. How much time does Kwame say elapses between events A and B?



Time

Time as a “fourth dimension” is unlike the three spatial dimensions

- “3+1”
- The arrow of time
- No “velocity” or “acceleration” of time -- requires another time dimension to have meaning

Most physical laws run the same forwards and backwards

- One possible cause: “Entropy”



Entropy As Time

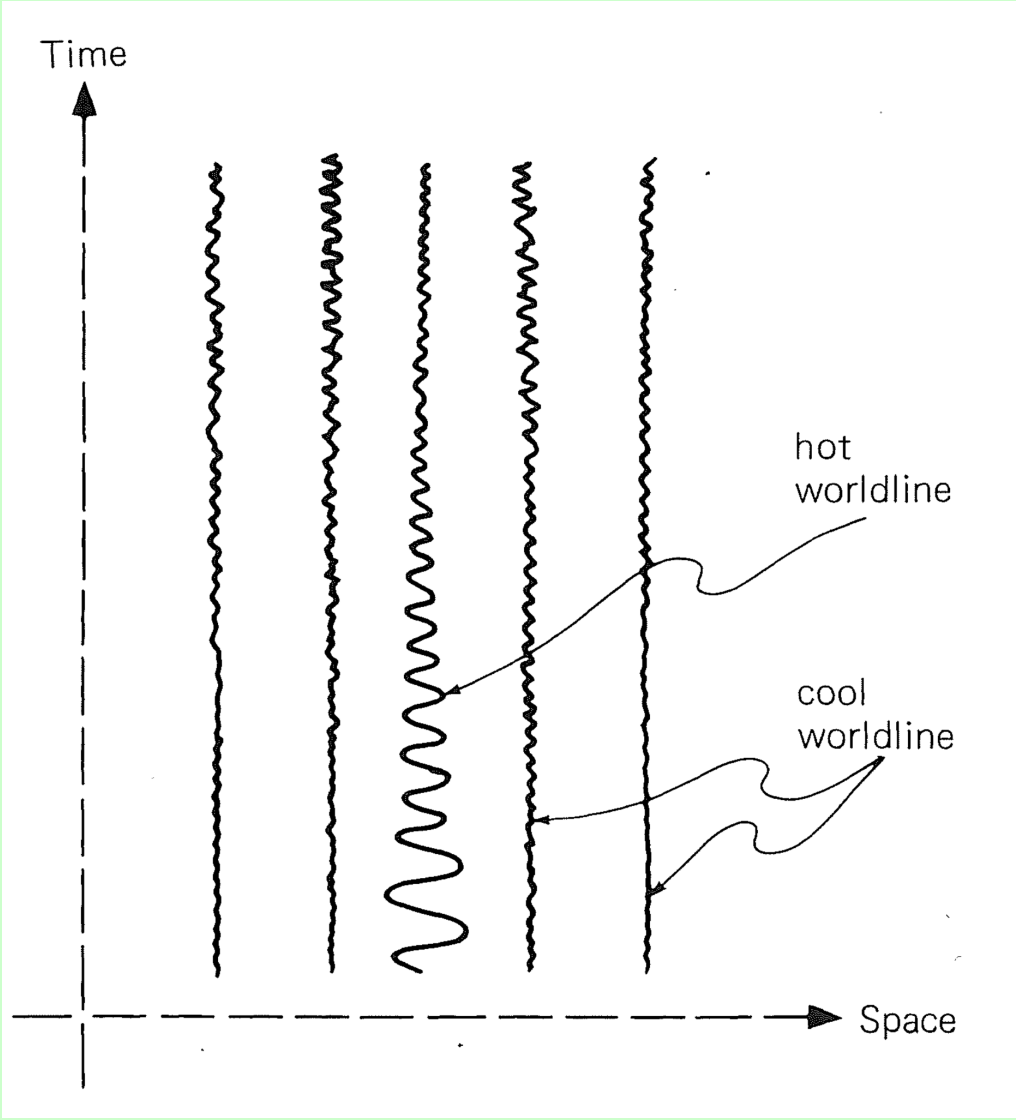
Question: What causes our perception of time's "arrow?"

Possible Answer: "Entropy"

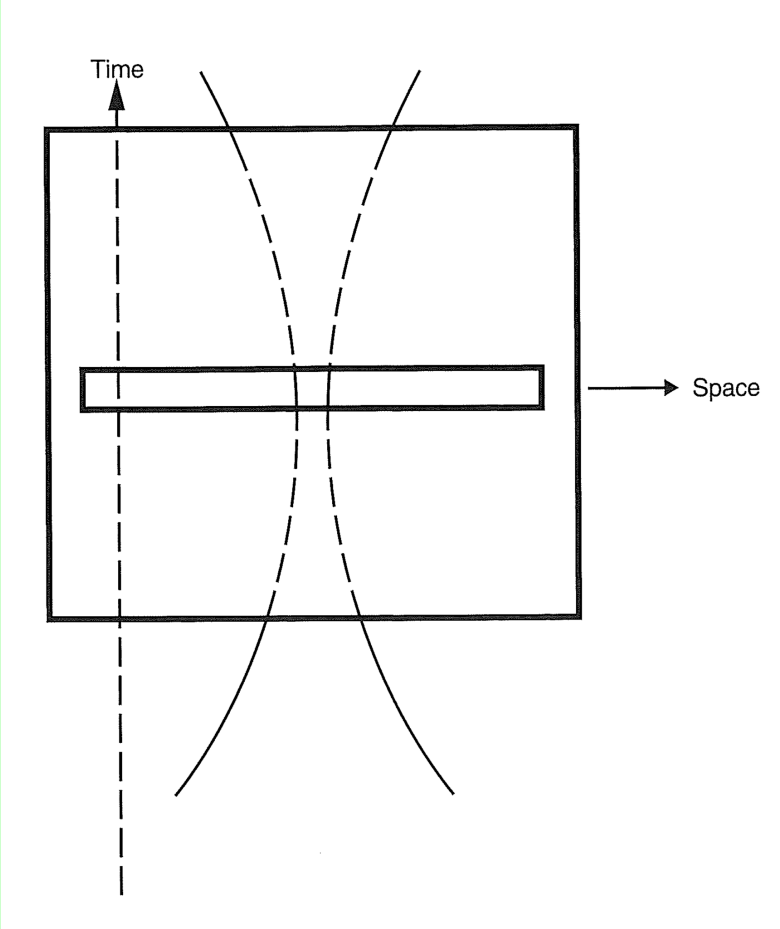
Worldlines of particles smooth out

That "is" why time appears to flow in one direction……

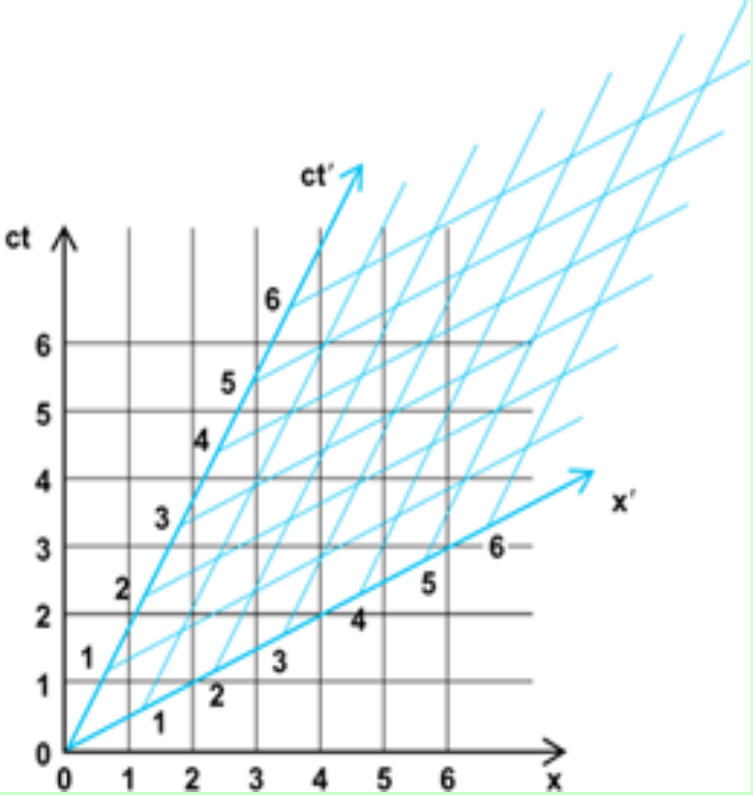
Does this lead to an end of time?



What is Now?

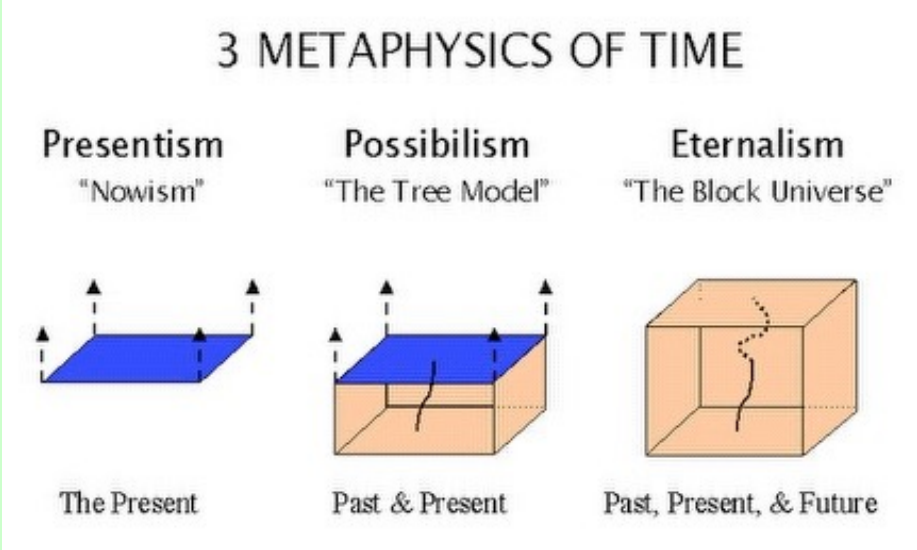


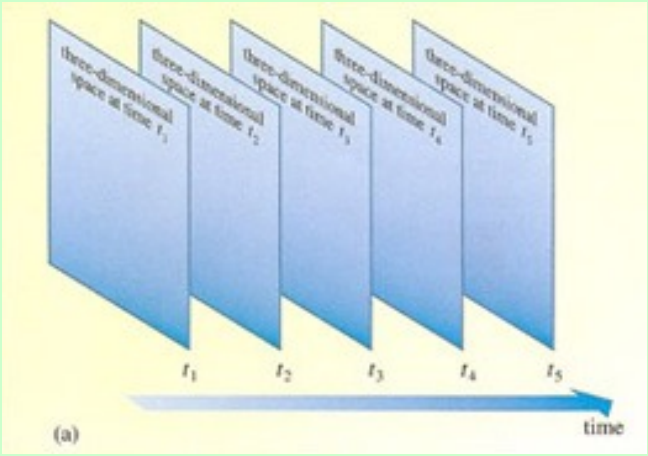
Cardbord Box View of “Now”



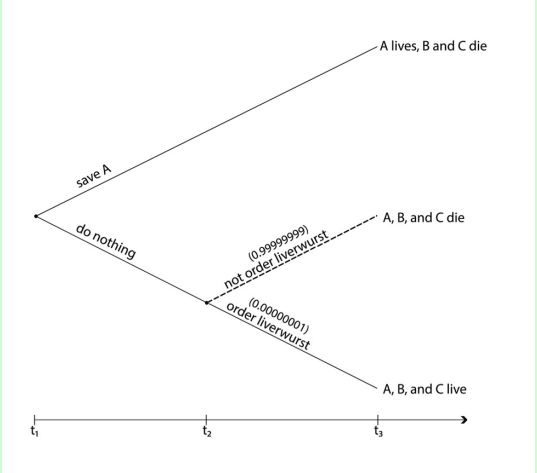
changes for somebody with
Motion relative to you…

Metaphysics of Time

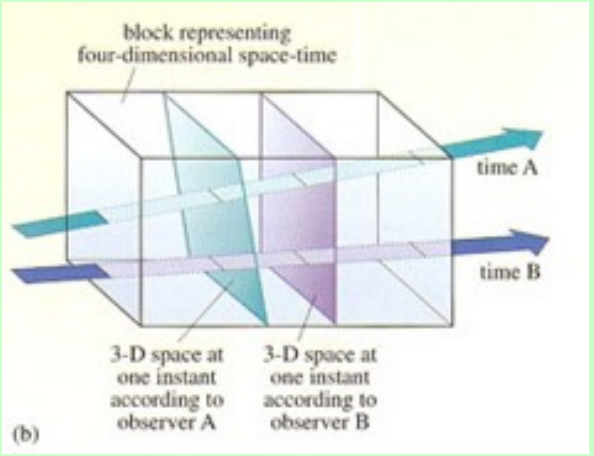




Nowism



Tree Model



Block Universe